SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV **COURSE CODE: DSA0202**

COURSE OUTCOME COVERED

CO2: *Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analysing images.. (BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 3 **Total Marks:30**

Annexure A

Scenario-Based

Code of Conduct:

I __P Sai Bhavyatha(192424333)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Palacharla Sai Bhavyatha*

RUBRICS FOR CALCULATION

Scenario based assessment						
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Marks**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	

Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Annexure A

Scenario-Based

1. Crack Detection on Concrete Surfaces Using Drone Images

Scenario: A civil inspection team is developing a computer vision system to detect thin cracks on concrete bridges and building surfaces. Images are captured using a drone under outdoor conditions. Due to changing sunlight, shadows, and camera vibration, the raw images contain uneven illumination, low contrast, and Gaussian noise. **Questions:**

A. Propose a suitable image preprocessing pipeline to prepare the drone images for crack detection. Your answer should include methods for illumination correction, noise reduction, contrast enhancement, and image normalization. Justify the need for each step.

B. Select an appropriate edge detection technique for identifying thin surface cracks. Compare it with at least one other edge detection method and explain why your selected technique is more suitable for detecting fine cracks.

C. After edge detection, some crack segments appear broken or disconnected. Suggest a suitable post-processing technique to connect the crack regions and improve the final detection output.

2. Human Motion Tracking in a Security Camera Video

Scenario: A security camera is installed in a hallway to monitor people moving in different directions. The system must estimate the motion of each person across video frames for tracking and activity analysis. The hallway contains plain walls, smooth floors, and occasional shadows.

Questions:

A. Explain how the Lucas-Kanade optical flow method can be used to estimate the motion of people between consecutive video frames.

B. Discuss how the aperture effect can create errors in motion estimation, especially on uniform regions such as plain walls, floors, and clothing surfaces.

C. If a person suddenly changes speed or direction, describe how the optical flow vectors will be affected. Suggest a method to improve tracking accuracy in such situations.

3. Vehicle Tracking at a Traffic Junction Using Video Frames

Scenario: A smart traffic monitoring system uses CCTV footage to track vehicles at a busy road junction. Vehicles move at different speeds, some stop suddenly at signals, and others turn left or right. The video also contains shadows, reflections, and partial occlusion when vehicles overlap.

Questions:

A. Explain how optical flow can be used to estimate the direction and speed of moving vehicles in the video sequence.

B. Identify two challenges that may affect accurate motion estimation in this scenario, such as occlusion, shadows, sudden braking, or camera vibration. Explain how each challenge affects the optical flow result.

C. Suggest a suitable improvement method to make vehicle tracking more reliable when vehicles overlap or change direction suddenly.

ANSWERS

Question 1: Crack Detection on Concrete Surfaces Using Drone Images

A. Image Preprocessing Pipeline

A suitable preprocessing pipeline for crack detection is:

1. **Image Acquisition**
   - Capture high-resolution images using a drone.
2. **Illumination Correction**
   - Use **Histogram Equalization** or **Adaptive Histogram Equalization (CLAHE)**.
   - **Reason:** Corrects uneven lighting caused by sunlight and shadows.
3. **Noise Reduction**
   - Apply a **Gaussian Filter**.
   - **Reason:** Removes Gaussian noise while preserving important image details.
4. **Contrast Enhancement**
   - Use **CLAHE** to improve the visibility of thin cracks.
   - **Reason:** Enhances local contrast and makes cracks more distinguishable.
5. **Image Normalization**

    - Normalize pixel intensity values to a fixed range (0–1 or 0–255).
    - **Reason:** Maintains consistent brightness and improves processing accuracy.

6. **Edge Detection**
    - Apply an edge detector to extract crack boundaries.

Justification:
Each preprocessing step improves image quality by reducing lighting variations, removing noise, enhancing crack visibility, and providing a standardized image for accurate crack detection.

B. Suitable Edge Detection Technique

The **Canny Edge Detector** is the most suitable method.

Comparison with Sobel Edge Detector

Feature	Canny	Sobel
Noise handling	Excellent	Moderate
Thin edge detection	Excellent	Good
False edge detection	Low	Higher
Edge localization	Accurate	Less accurate

Why Canny?

- Detects very thin cracks accurately.
- Uses Gaussian smoothing to reduce noise.
- Produces continuous and well-defined edges.
- Minimizes false edge detection.

Therefore, **Canny** is preferred over Sobel for crack detection.

C. Post-Processing Technique

Use **Morphological Closing** (Dilation followed by Erosion).

Advantages

- Connects broken crack segments.

- Fills small gaps.
- Produces continuous crack structures.
- Improves crack detection accuracy.

Question 2: Human Motion Tracking in a Security Camera Video

A. Lucas-Kanade Optical Flow

Lucas-Kanade estimates the motion of feature points between two consecutive video frames.

Working:

1. Detect feature points.
2. Assume motion is small between frames.
3. Compute pixel displacement using neighboring pixels.
4. Generate optical flow vectors showing motion direction and speed.

This helps track people moving through the hallway.

B. Aperture Effect

The aperture effect occurs when motion is observed only through a small region.

In plain walls, smooth floors, or clothing:

- Very few texture features are available.
- Motion direction becomes ambiguous.
- Optical flow vectors become inaccurate.
- Tracking performance decreases.

Therefore, textured regions provide more reliable motion estimation.

C. Sudden Change in Speed or Direction

If a person suddenly changes speed or direction:

- Optical flow vectors change rapidly.
- Motion estimation may become inaccurate.
- Tracking may temporarily fail.

Improvement Method

Use the **Kalman Filter**.

Benefits

- Predicts future object position.
- Handles sudden motion changes.
- Reduces tracking errors.
- Improves tracking stability.

Question 3: Vehicle Tracking at a Traffic Junction

A. Optical Flow for Vehicle Tracking

Optical flow estimates the movement of vehicles by comparing consecutive video frames.

It provides:

- **Direction of movement** (left, right, forward).
- **Speed** (based on pixel displacement).
- **Motion vectors** for each moving vehicle.

These vectors help continuously track vehicles through the traffic junction.

B. Challenges Affecting Motion Estimation

1. Occlusion

When vehicles overlap:

- Some vehicles become partially hidden.
- Motion vectors become incomplete.
- Tracking accuracy decreases.

2. Shadows

Vehicle shadows also move.

Effects:

- Shadows may be detected as moving objects.
- False motion vectors are generated.
- Vehicle boundaries become unclear.

These factors reduce the accuracy of optical flow estimation.

C. Improvement Method

Use **Kalman Filter with Object Detection (e.g., YOLO)**.

Advantages

- Predicts vehicle positions during temporary occlusion.
- Continues tracking even when vehicles overlap.
- Handles sudden turns and braking.
- Improves overall tracking accuracy.
- Reduces tracking loss and false detections.

Conclusion

The proposed methods improve the performance of computer vision systems by:

- Enhancing image quality through preprocessing.
- Accurately detecting edges using the **Canny Edge Detector**.
- Improving crack connectivity with **Morphological Closing**.
- Estimating motion using **Lucas-Kanade Optical Flow**.
- Handling tracking challenges with the **Kalman Filter** and object detection techniques.

These approaches provide reliable and accurate solutions for crack detection, human motion tracking, and vehicle tracking in real-world applications.